

AISSAOUI Ismail

State Engineer in Artificial Intelligence & Data Science

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PROFESSIONAL SUMMARY

State Engineer graduated from ESI-SBA, specializing in Transformers, NLP, Digital Twins, and Industrial AI. Expert in designing end-to-end AI solutions, from metaheuristic optimization to MLOps deployment. PhD candidate focused on transforming industrial systems through hybrid AI and cyber-physical modeling.

WORK EXPERIENCE

AI Engineer Intern — Generative Digital Twin Architecture Jan 2026 – Present
Sonatrach (Pipeline Transport Branch)

- Designed a Generative Surrogate based on **Mamba-SSM** for real-time "What-If" simulations.
- Hardware-Software Co-Design: **0.04ms** edge inference via **C++ JIT Compiled xLSTM**.
- Integrated **PINN** (Physics-Informed Neural Networks) to ensure thermodynamic monotonicity.

Network Engineering Intern Sep 2024
Algérie Télécom RMC Center

EDUCATION

National School of Computer Science (ESI-SBA, Algeria) 2021 – 2026
State Engineer's Degree & MSc in Artificial Intelligence

TECHNICAL SKILLS

- AI / Deep Learning:** PyTorch, TensorFlow, Transformers, GNN, GANs, RL (DQN/PPO).
- Optimization:** PSO, Genetic Algorithms, ACO, Simulated Annealing, Bayesian Optimization.
- Tech Stack:** Python, FastAPI, Next.js, Docker, MLflow, Kubernetes, CUDA, SQL.

RESEARCH & ENGINEERING PROJECTS

Aura — Adaptive Reinforcement Learning System 2025
Institutional-grade trading platform (LSTM + DQN) with real-time WebSocket dashboard.

Career Connect — GNN recruitment Intelligence 2024 – 2025
Complex semantic matching using Graph Neural Networks and relational embeddings.

Advanced NLP Optimization (Metaheuristics) 2025
Multilingual moderation optimized via PSO, GA, and Bayesian optimization (DistilBERT).

University Scheduling (NP-Complete Optimization) 2025
Automated timetabling via hybrid heuristics (PSO/ACO/GA) for conflict-free allocation.

Mini-GPT & North African Sentiment BERT 2025
From-scratch Transformer implementation and LoRA adaptation for Maghrebi dialects.

Geo-RAG — Intelligent Map Assistant 2025
Geospatial RAG system integrating ChromaDB and GeoPandas for spatial reasoning.

Recognizini — Interactive Facial Recognition 2025
Online learning system with incremental embeddings and Active Learning feedback loops.

Computer Vision & Industrial Safety (YOLOv8) 2025
Traffic analytics (Tracking/OCR) and real-time PPE (Safety Equipment) detection.

Synergy POS & Hadj Logistics 2024 – 2025
Complex logistics and retail solutions (Offline/Online Sync, PostgreSQL/SQLite).

LANGUAGES

Arabic (Native), English (Advanced), French (Fluent).